Section 3: EvalzNow Fundamentals

Before starting, it is necessary to define the meaning of the word, “assessment”, associated with human capital and knowledge capacity because this word is oftentimes confused with examinations, evaluations, and surveys in education and industry. An assessment, by definition, is the systematic process of defining, selecting, designing, collecting, analyzing, interpreting, and monitoring data to understand the learning capacity and the knowledge development of a candidate. Contrary to popular belief, an assessment requires qualitative data and quantitative data to fulfill all avenues of the process. Other notes are provided below.

- An examination takes place after instruction/training to determine if an objective has been reached; however, an assessment takes place before/during/after instruction/training to determine how effective the instruction/training was at reaching an objective.
- An evaluation, which is not seamless, uses restricted, short-listed criteria to validate a candidate’s value, quality, significance, grading, and ability against a pre-defined standard at a particular time. An assessment, on the other hand, is a comprehensive process that goes well beyond the evaluation standard to make records of a candidate’s knowledge capacity holistically.
- A survey is used to gather data, which might or might not be reliable, at an elementary level for a particular purpose; however, an assessment juxtaposes qualitative data and quantitative data based on primary sources and direct observations.
- Since the assessment process is detailed and it is highly extensive, it is normally broken into smaller components. For instance, market assessments, job assessments, and operational assessments are smaller components of an industry assessment, while diagnostic assessments, formative assessments, and year-end assessments are smaller components of an educational assessment. Also, examinations, evaluations, and surveys can be integrated into assessments, but not the other way around.

All qualitative aspects and all quantitative aspects of all assessment outcomes must be absorbed, and they must be calculated in a centralized, evidence-based system to determine the knowledge capacity of a candidate at a detailed level spanning long periods. EvalzNow is such a system.

EvalzNow provides detailed, human capital assessments using an innovative approach known as Competency Accounting. As the approach suggests, Competency Accounting is the measurement, the processing, and the communication of quantitative information and qualitative information regarding any competency-based entity, such as students, employees, AI humanoids, educational facilities, businesses, defense agencies, and so forth. The foundation of the Competency Accounting approach centers on grouping independent variables pertaining to the cerebral structure into exclusive categories prior to data processing, similar to business accounting. In business accounting, for instance, the independent variables, debits and credits, are grouped separately before they are processed into subcategories of assets (accounts receivable, trademarks, etc.), liabilities (accounts payable, taxes, etc.), and equity (shareholder, stock, etc.). Analogously, in Competency Accounting, the independent variables are processed into subcategories based on the models below.

Communicative I/O Model

The communicative inputs are the competencies, listening and reading.

The communicative outputs are the competencies, speaking and writing.

This model has applicability to any subject that captures listening/reading inputs and speaking/writing outputs. Notable examples include language/linguistics, literature, humanities, social studies, civics, history, psychology, philosophy, psychology, geography, health, and sociology.

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Theory & Practice Model

The competencies are theoretical inputs that involve reasoning, techniques, fundamental concepts, critical thinking, brainstorming, planning, theory, and qualitative procedures.

The competencies are practical outputs that involve hands-on activities, lab work, application, testing, design, implementation, processes, and quantitative procedures.

This model applies to any subject that captures theoretical inputs and practical outputs. Notable examples include corporate entities, geometry, algebra, calculus, linear algebra, PDE/ODE, numerical methods, chemistry, physics, engineering, economics, woodshop, programming, instrumentation, music, and medicine.

Theoretical knowledge, by definition, refers to knowledge accumulated from studies or explanations based on presumptive ideas and abstract principles about a particular subject.

Practical knowledge, by definition, refers to knowledge accumulated from real-life applications, witnessed events/situations, and hands-on experience.

Third Parameter Model

The first competency inputs are the receptive variables.

The second competency inputs are the acuity variables (fitness is also applicable).

The competency outputs are the productive variables.

This model, which represents the general case, applies to defense, gifted & talented, deep learning, sports, etc.

Evaluations of visual acuity, auditory acuity, mental acuity, and mental/physical fitness, which contribute to acuity variables, involve comparing specimen performance test data against empirical baselines or threshold parameters considered to be normal for the average human being. Bear in mind that counterintelligence assessments, deep/forensic analysis, critical reasoning, special operations, and various deep psychometric assessments lie within the scope of acuity variables so long as the quantifiers measure the sharpness of thought, vision, hearing, and/or intellect (E.G. IQ).

Regardless of the accounting system, the intent of accounting is to provide a microscopic view (detailed view) and a macroscopic view (global view) of data generated by a given entity. Such views cannot be achieved, however, if there is not enough fidelity in the recordkeeping process. In educational entities (K-12 schools, universities, etc.), corporate entities, and defense entities, grading, ranking, and performance measures are based on lumped data records; therefore, there is no way to assess proficiency & growth or to isolate all problematic areas at the candidate level (students, employees, etc.) and the unit level (school, company, etc.). Reverting to business accounting as an analog once more to prove a point, lumped data records would result in a company having one (1) economic metric, profit, rather than balance sheets, income statements, cash flow summaries, and financial ratios. Clearly, there are accountants to solve these issues on the business side; however, what about the human capital side? EvalzNow reigns solely on the human capital side because it is derived from a sound methodology that does not lump human capital data for the first time. As a result, the user can track hybrid, proficiency and growth, metrics alongside observations and tertiary factors across large periods.

The Communicative I/O Model is compatible with Butler Signature Cases 1-3b.

The Theory & Practice Model is compatible with Butler Signature Cases 1-3b.

The Third Parameter Model is compatible with Butler Signature Cases 1.

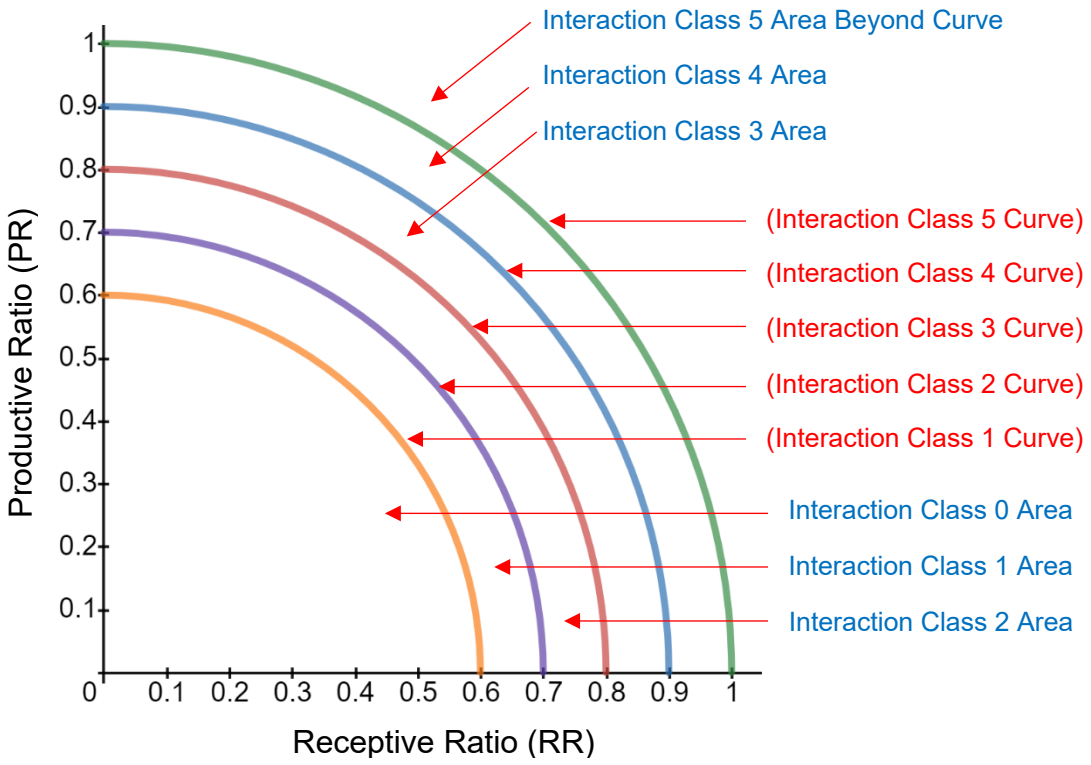
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BUTLER SIGNATURE CASE 1: SOLID AND BALANCED INTERACTION CURVES

The intent of this case is to balance the input/output variables alongside attainable growth targets. This case should be the standard default for core courses of K-12 schools, international frameworks, deep assessments, and industry.

The input variables are quantified as a receptive ratio (RR).

The output variables are quantified as a productive ratio (PR).



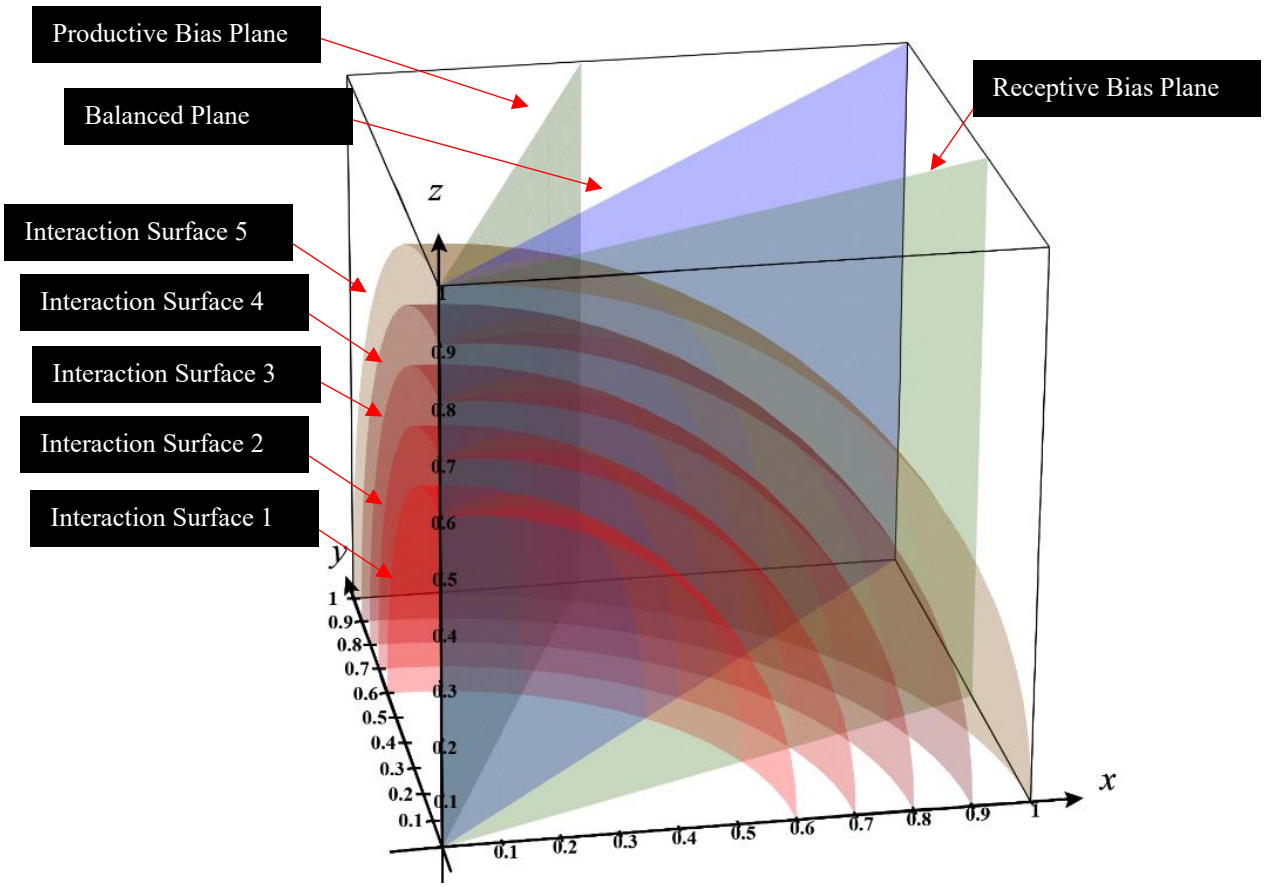
If the user wants to add the third parameter to the assessment process, then the acuity variable must be considered. The intent is to balance the receptive variables, the productive variables, and the acuity variables alongside attainable growth targets. This case should be the standard default for military, deep learning, advanced psychometrics, gifted & talented assessments, etc.

The receptive variables are quantified as a receptive ratio (RR) along the X-axis.

The productive variables are quantified as a productive ratio (PR) along the Y-axis.

The acuity variables are quantified as an acuity ratio (PR) along the Z-axis.

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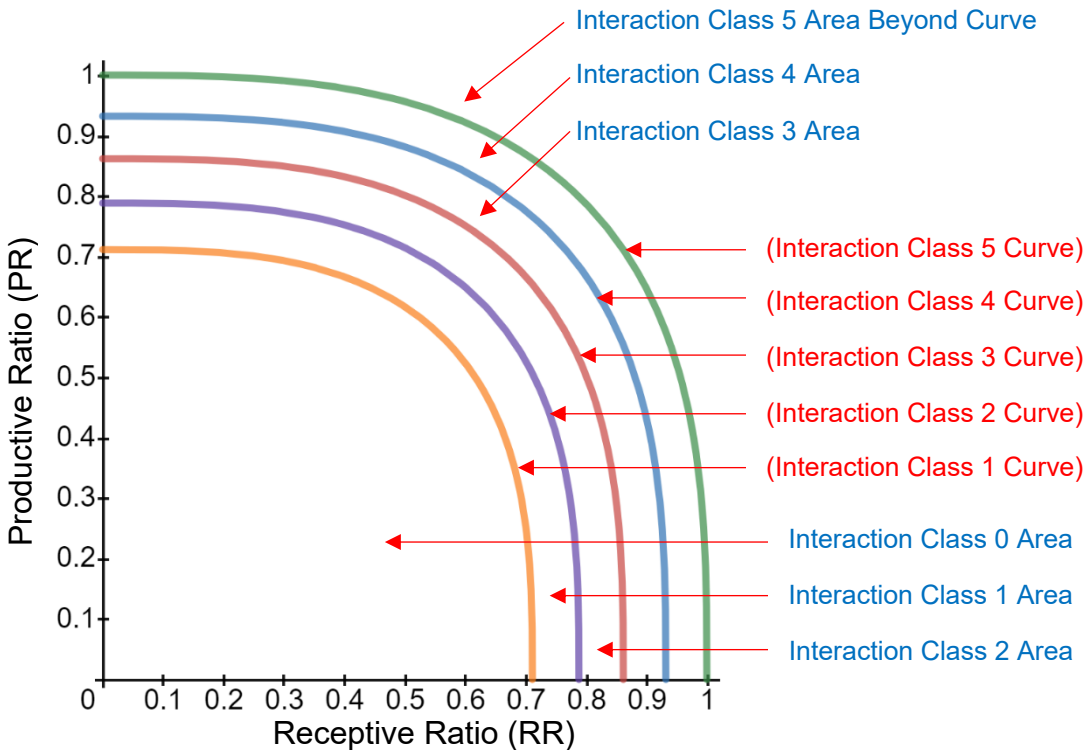
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BUTLER SIGNATURE CASE 2: ROBUST AND BALANCED INTERACTION CURVES

The intent of this case is to balance the input/output variables alongside challenging growth targets. This case should be the default with Advanced Placement (AP) courses, International Baccalaureate (IB) courses, accelerated/intensive frameworks, and Career Programs (CP). Different industries can also use this case to speed-track employees for highly competitive positions.

The input variables are quantified as a receptive ratio (RR).

The output variables are quantified as a productive ratio (PR).



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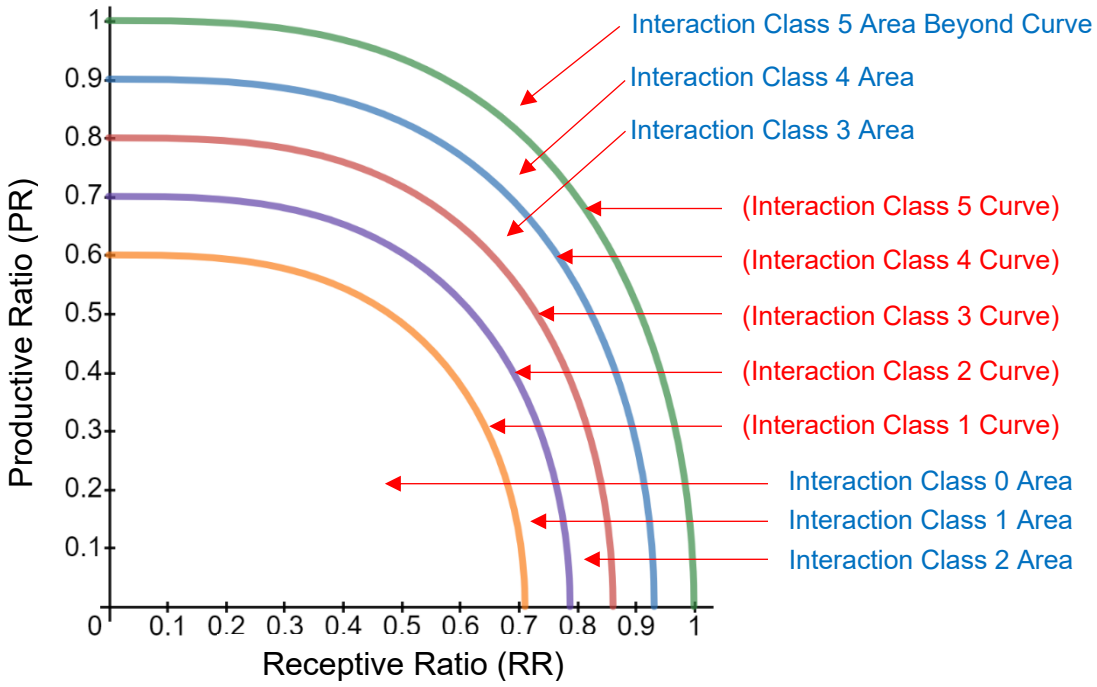
BUTLER SIGNATURE CASE 3: ROBUST & IMBALANCED INTERACTION CURVES

The intent of this case is to weigh the output variables differently from the input variables. This case is recommended for conversational courses, speed writing courses, fine arts, and so forth, where the emphasis is either on input variables or output variables.

The input variables are quantified as a receptive ratio (RR).

The output variables are quantified as a productive ratio (PR).

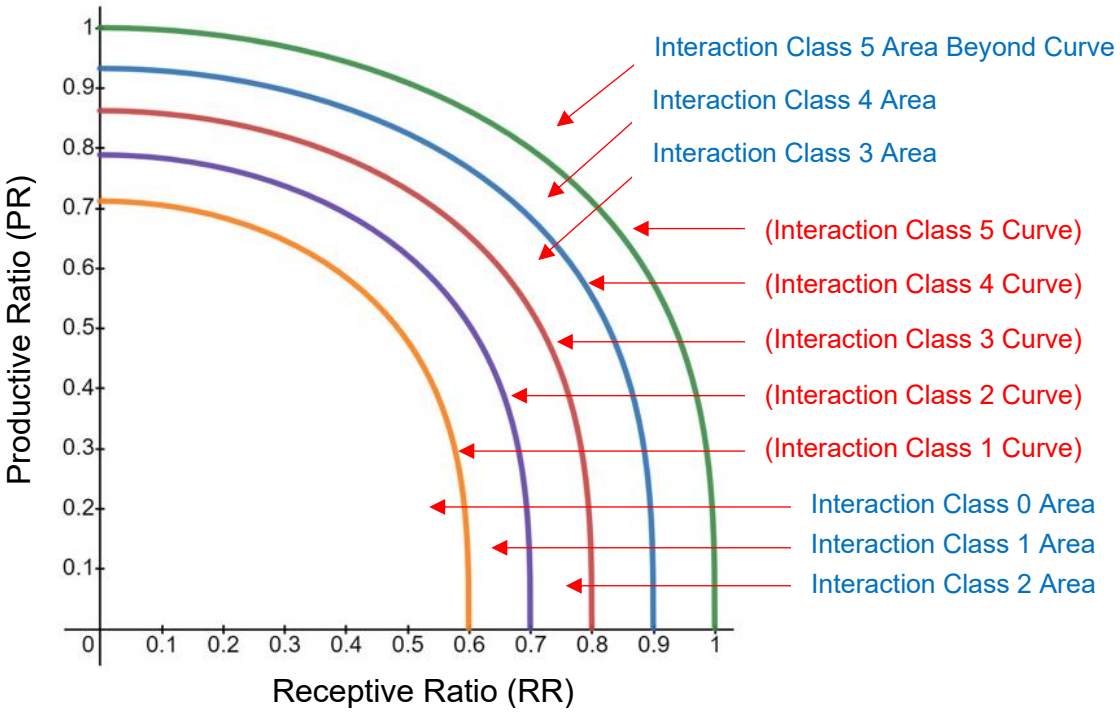
Case 3a: Weighting of Productive Variables



It is worth noting that the productive ratio of the interaction class curves is drawn closer to the unit square boundary than the receptive ratio of the interaction class curves. This behavior is attributed to the weight imbalance of the interaction curves that drift toward the output variables.

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Case 3b: Weighting of Receptive Variables



It is worth noting that the receptive ratio of the interaction class curves is drawn closer to the unit square boundary than the productive ratio of the interaction class curves. This behavior is attributed to the weight imbalance of the interaction curves that drift toward the input variables.

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HYBRID, PROFICIENCY AND GROWTH, ASSESSMENT METRICS

Growth in the learning process refers to the differential changes in input variables and in output variables from one performance point to another performance point. These differential changes, which can be correlated with normative measures at a student level, a classroom level, or a corporate level, indicate the direction of individual progress. It is also a factor for instructor accountability.

If the differential change in input variables and/or output variables is negative, then the direction of student/employee progress decreases due to retarded growth.

If the differential change in input variables and/or output variables is zero, then the direction of student/employee progress remains unchanged with no growth.

If the differential change in input variables and/or output variables is positive, then the direction of student/employee progress increases due to value-added growth.

Please note that EvalzNow incorporates proprietary growth projection factors to determine the average projected performance points before sending them to the Butler Hybrid Model. All growth projection calls should be sequential, they should not be repeated/duplicated, and they should have no missing performance point data (I.E. the receptive ratios, the productive ratios, and the acuity ratios should be known for each PP selection.)

Please note that EvalzNow automatically calculates growth targets for all performance points subject to complex and proprietary calculations. These growth targets are provided in tabular form and graphical form.

A methodology for determining a maximum of five (5) growth targets that lead to value-added growth and increasing student/employee progress across performance points (PP) has been integrated into EvalzNow.

PP → Growth Target 1 → Growth Target 2 → Growth Target 3 → Growth Target 4 → Growth Target 5

The number of growth targets depends on the performance point, and growth targets become more aggressive in succession.

Student growth from one performance point to another performance point is evaluated based on the following two (2) criteria:

1. The performance point corresponds to an interaction class greater than or equal to Interaction Class 2.

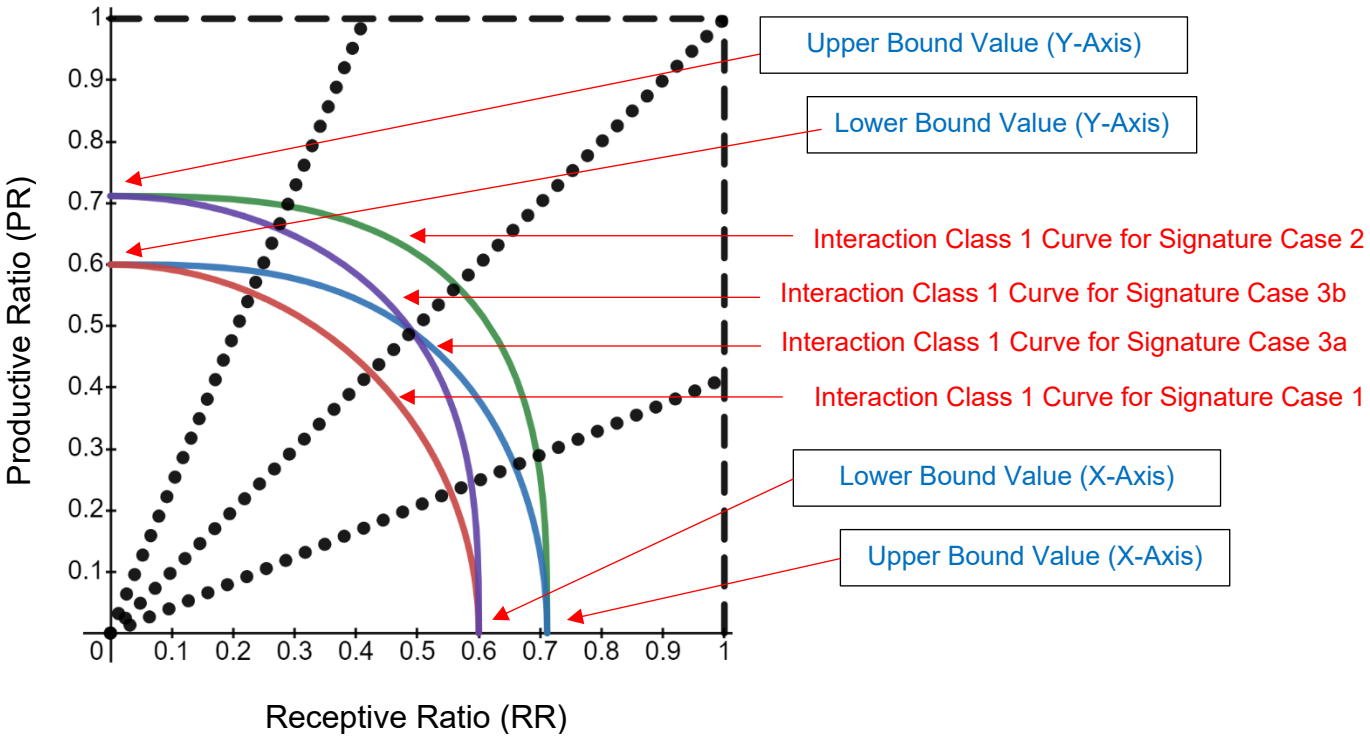
The interaction class of a performance point depends on where norm PP = < RR, PR, AR > falls within the boundaries of the individual interaction class curves.

- Interaction Class 0 through Interaction Class 5 Areas for Signature Case 1.
- Interaction Class 0 through Interaction Class 5 Areas for Signature Case 2.
- Interaction Class 0 through Interaction Class 5 Areas for Signature Case 3a or 3b.

→ Interaction Class 0 refers to the **failing** interaction class.
→ Interaction Class 1 refers to the **at-risk** interaction class.
→ Interaction Class 2 refers to the **acceptable** interaction class.
→ Interaction Class 3 refers to the **standard** interaction class.
→ Interaction Class 4 refers to the **advanced** interaction class.
→ Interaction Class 5 refers to the **exceptional** interaction class.

Remember that the purpose of the interaction class is to identify underperforming candidates, at-level candidates, and overperforming candidates by creating categorical ranges from “failing” to “exceptional”. To ensure that candidates are not misclassified, EvalzNow calculates the interaction class value using the Interaction Class Curves from Signature Case 1. This calculation provides flexible, yet feasible, upper/lower bounds, which are shown in the figure below by juxtaposing Interaction Class 1 Curves for all Signature Cases.

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Recall that the four (4) signature cases – Signature Case 1/2/3a/3b – are collectively applicable to the Communicative I/O Model and the Theory & Practice Model. Drawing attention to Interaction Class Curve 1 only, the flexible upper/lower bounds of each Signature Case for each model are as follows:

- Interaction Class 1 (Signature Case 1): Lower Bound → 0.6000, Upper Bound → 0.6000
- Interaction Class 1 (Signature Case 2): Lower Bound → 0. 7114, Upper Bound → 0.7114
- Interaction Class 1 (Signature Case 3a): Lower Bound → 0. 6000, Upper Bound → 0.7114
- Interaction Class 1 (Signature Case 3b): Lower Bound → 0.6000, Upper Bound → 0.7114

The values of the lower bounds and the upper bounds prove that Signature Case 1 is the pragmatic case for creating categorical ranges, especially since this case can accommodate companies with strict fairness/appeal policies. The user has full discretion in using the lower bound, the upper bound, or a value between the lower bound and the upper bound in the decision-making process (E.G. utilizing the Butler Hybrid Model directly); however, EvalzNow will always take the lower bound of Signature Case 1.

The procedure for determining the lower/upper bounds is the same for Interaction Class Curve 2-5 for the Communicative I/O Model and the Theory & Practice Model. For the Third Parameter Model, however, the results can be used 'as is' because the rigorous surface model is an extension of Signature Case 1.

2. The receptive growth differential and the productive growth differential among consecutive performance points must be positive.

$$\Delta PP = < \Delta RR, \Delta PR, \Delta AR >$$

- If $\Delta RR > 0$ for consecutive performance points, then the direction of student progress increases due to **value-added growth for receptive variables**.
- If $\Delta PR > 0$ for consecutive performance points, then the direction of student progress increases due to **value-added growth for productive variables**.
- If $\Delta AR > 0$ for consecutive performance points, then the direction of student progress increases due to **value-added growth for acuity variables**.

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The window for student growth shrinks as the student performs better from one (1) performance point to another performance point.

Unlike student growth, which encourages accountability on behalf of instructors, proficiency passes the accountability to the student. Proficiency, which is not a reflection of student progress, is a metric that is used to determine if a student can meet or can exceed a predefined benchmark or a baseline standard.

In the student learning process, there are four (4) different categories for measuring student proficiency.

- **Unilateral Proficiency** is used to determine if a student can demonstrate mastery of individual skills based on a predefined benchmark or a baseline standard.

- **Composite Proficiency** is used to determine if a student can demonstrate mastery across all skills based on a predefined benchmark or a baseline standard.

- **Band Proficiency** is used to determine if a student can demonstrate mastery by averaging scores across all skills for comparison against a predefined benchmark or a baseline standard.

- **Butler Interaction Proficiency** is used to determine if a student can demonstrate mastery across all skills based on an acceptable interaction among receptive variables, productive variables, and acuity variables.

Please note that the Butler Interaction Proficiency is the only proficiency metric that should fall below the user threshold value because lower values indicate that the candidate has exceeded the knowledge envelope.

Please note that the user should never crowd a performance point by lumping a large number of assignments. As soon as all competencies of the desired mathematical model are cleared for a performance point, open another performance point. Lumping a large number of assignments into a performance point decreases the fidelity of the data inputs/outputs, the feasibility of the critical reference date, and the statistical reliability.

Please note that the stability of the growth and the proficiency metrics are provided in the software.

CRITICAL REFERENCE DATES

EvalzNow relies on accurate time stamps to deliver progressive data analytics based on logged tasks. This means that all tasks contained within assessment blocks must be sequential, and all assessment blocks must be sequential, subject to -/+ performance points created in the assessment block delivery.

Once again, the purpose of critical reference dates is to keep all logged tasks, as well as their scores and analytics, in order with respect to time. The system is sensitive to user inputs, especially for new entries, because it assembles, sorts, and calculates an entire matrix based on multiple tasks contained within a performance point.

As a general rule, never pack a performance point with a large number of logged tasks because the fidelity decreases in the results. The recommended number of maximum logged tasks within a performance point is six (6) for the Communicative I/O Model, it is four (4) for the Theory & Practice Model, and it is five (5) for the Third Parameter Model. The user can also choose to open another performance point after all competencies of a given model have been covered by the logged tasks.

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UNIT DISTRIBUTIONS

EvalzNow integrates statistical distributions to assess candidate data pools. These distributions include the normal distributions, exponential distributions, chi-square distributions, F distributions, Weibull distributions, and beta/uniform distributions.

1. **Normal Distribution**: The normal distribution is a theoretical distribution of values for a given sample size. Often referred to as a bell curve when plotted on a graph, sample data with a normal distribution tends to accumulate around a central value (mean at $\sigma=0$); the frequency of values above and below the center declines symmetrically. For EvalzNow, the normal distribution is provided for the following sample data: Normalized Receptive Ratios (RRN), Normalized Productive Ratios (PRN), Normalized Acuity Ratios (ARN), Equivalent Proficiency, Composite Proficiency, Band Proficiency, Receptive Growth, Productive Growth, Acuity Growth, and Equivalent Growth.
2. **Exponential Distribution**: The exponential distribution is a right-skewed continuous probability distribution that models variables in which small values occur more frequently than higher values across the timing between independent events. Please keep in mind that the time for the next event does not depend on how much time has elapsed, and the likelihood of when the event occurs next does not depend on when it happened previously; thus, the constant average rate of a sample size is utilized to visualize the decay. For EvalzNow, the exponential distribution is provided as a function of time across a 15-day period for the following sample data: Normalized Receptive Ratios (RRN), Normalized Productive Ratios (PRN), Normalized Acuity Ratios (ARN), Equivalent Proficiency, and Equivalent Growth (ΔGEQ). Please note that this distribution is valid only for positive data.
3. **Chi-Square Distribution**: The chi-square distribution determines whether there are statistically significant differences between expected sample data and observed sample data. For EvalzNow, the average proficiency and growth metrics are tested against threshold proficiency and growth metrics to ensure that there is an acceptable correlation between proficiency and growth, as well as an acceptable correlation between the interaction of proficiency and growth. To be accepted for a given data sample, the Chi-Square value must be less than the critical value (Xc).
4. **F-Distribution**: The F-Distribution is a hypothesis test based on the ratio of variances from two (2) data samples when the samples are drawn independently from normally distributed samples. For EvalzNow, the two (2) data samples are the proficiency metrics and the growth metrics across a range of independently drawn samples. Therefore, Fstat should also exceed the critical value Fc because there is never equality of the two (2) variances.
5. **Beta Distribution**: The Beta Distribution is used to model random data samples as binaries taken from data samples with distinguishable characteristics. For EvalzNow, the binaries of the sample data, as well as the shape parameters, are taken as 'stable' and 'unstable' events to determine the probability or the likelihood of always having a stable interaction of proficiency and growth (mutual causality).
6. **Uniform Distribution**: The Uniform Distribution is used to show the trends between binary states ('Stable' states and 'Unstable' states in the case of EvalzNow) across all sample data.
7. **Weibull Distribution**: The Weibull Distribution is a continuous probability distribution used to determine the behavior and the failure pattern of sample data for a given set of parameters – mostly the scale parameter η and the shape parameter β. For EvalzNow, the scale parameter is the characteristic value taken as the average of the sample data, and this value can be juxtaposed against the thresholds of a given metric. The shape parameter β, however, can indicate early failure of the sample data ($\beta < 1$), random failures of the sample data ($\beta = 1$), or wear-out failures of the sample data ($\beta > 1$). The user is free to investigate the failure envelopes; however, the recommended shape parameters are always $\beta = 0.5$, $\beta = 1.0$, and $\beta = 1.5$ for EvalzNow metrics, which look into underperformers, benchmark failures, framework flaws, proficiency/growth planning, or any other objective. EvalzNow provides the cumulative distribution function, the reliability function, the probability density function, and the risk/hazard function. The distribution is only valid for positive data.

EVALZNOW PROCESS

- The Candidate Level tracks individual candidates under assessment across unlimited date ranges.
- The Unit Level tracks each unit under assessment across unlimited date ranges.
- The Global Level tracks the institution of the client across unlimited date ranges.

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